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Enhanced photovoltaic cooling using sorption material with comparative evaluation of different fin geometries: A computational approach

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ABSTRACT

Enhancing the cooling efficiency of photovoltaic thermal (PVT) systems is essential for improving energy conversion in high-temperature environments. This study presents a computational approach using COMSOL Multiphysics to evaluate the integration of a composite sorption material (Silica Gel–$CaCl_2$, 3 cm thick) with five fin geometries (rectangular, horizontal cylindrical, triangular, trapezoidal, and T-shaped) for PV cooling. Simulations were performed under real climate conditions in Cairo, Egypt (30.02°N, 31.13°E) on July 12, 2024, when peak ambient temperature reached 43.58 °C and solar irradiance peaked at 976.33 W/m^2. Results show that the uncooled PV module reached a maximum temperature of 66.9 °C, while the addition of sorption material alone reduced this to 56.0 °C. The inclusion of fins further enhanced cooling, with rectangular fins lowering the peak temperature to 38.6 °C, representing a 28.3 °C reduction. Average PV module temperature decreased from 54.01 °C (uncooled) to 37.75 °C (rectangular fins). Efficiency analysis indicated an improvement from 13.03 % (uncooled) to 13.78 % (sorption only), with the highest efficiency of 14.13 % achieved by rectangular fins. Uptake performance confirmed superior thermal regulation, with rectangular fins reaching the highest value (0.346) compared to 0.302 for the sorption-only case. These findings demonstrate that coupling sorption materials with comparative evaluation of different fin geometries significantly enhances PV cooling and provides actionable guidelines for improving solar energy efficiency in hot-arid climates.

1. Introduction

Photovoltaic thermal systems (PVT) are a key technology in renewable energy generation, but their efficiency is significantly affected by high operating temperatures. To address this challenge, researchers have explored various cooling and thermal management techniques to enhance both electrical and thermal performance. Passive cooling methods, such as natural convection and heat sinks, are simple but less effective than active cooling techniques. Active cooling, including water and air-based systems, has shown significant temperature reductions (up to 18 °C) and efficiency improvements (1–18 %) [1]. Hybrid cooling methods, such as combining nanofluids with microchannel heat sinks, further enhance heat dissipation [2]. Phase change materials (PCM) are increasingly used in PVT systems due to their high latent heat storage capacity [3]. Studies indicate that PCM integration can reduce Photovoltaic (PV) temperatures by up to 25 K and improve electrical efficiency by over 10 % [4]. However, the low thermal conductivity of PCMs remains a limitation. To overcome this, researchers have incorporated nanoparticles, porous matrices, and fins to enhance heat transfer [5]. Despite progress, challenges remain in material durability, cost-effectiveness, and large-scale implementation [6].

Ibrahim et al. [7] demonstrated that PCMs improve PV efficiency by 4.23 % while reducing CO_2 emissions by 367.92 kg annually in domestic applications. Similarly, Yuan et al. [8] integrated PCMs with water cooling in dual-channel systems, achieving a 63.33 % overall efficiency. Challenges such as PCM leakage were addressed by *Al* Miaari [9], who developed a shape-stabilized PCM (SSPCM) using oil ash and RT-42, reducing PV temperature by 6.1 °C and enhancing efficiency by 9.02 %. Hygroscopic hydrogels offer evaporative cooling through moisture absorption-desorption cycles. Lu Huang et al. [10] achieved an 11.7 °C temperature reduction using hydrogel-integrated PVC-TEG devices,

Abbreviations: PV, Photovoltaic; PVT, Photovoltaic Thermal; PCM, Phase Change Material; SSPCM, Shape-Stabilized Phase Change Material; PVC-TEG, Photovoltaic Cooling-Thermoelectric Generator; TIM, Thermal Interface Material; PGS, Pyrolytic Graphite Sheet; HDH, Humidifier-Dehumidifier; CFD, Computational Fluid Dynamics; CF-AHE, Copper Foam-Absorbent Hydrogel Evaporative; PV/HG, Photovoltaic/Hydrogel; Ag-water, Silver-water nanofluid; CuO-water, Copper Oxide-water nanofluid; AHE, Absorbent Hydrogel Evaporative; CF, Copper Foam; TEG, Thermoelectric Generator; MOF, Metal–Organic Frameworks.

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Nomenclatures

A	Area of PV panel, m^2
$C_{p,sorp}$	Sorption material specific heat capacity of, J/K.kg
$C_{p,water}$	Water specific heat capacity, J/K.kg
D_{sdo}	Surface diffusivity factor, m^2/s
E_{act}	Activation energy, J/mol
E_c	The characteristic energy, J/mol
H_s	Isosteric heat of adsorption of the sorption material, kJ/kg
h_{bc}	Convective heat transfer coefficient of the sorption material bottom side, W/K.m^2
h_{br}	Radiation heat transfer coefficient of the sorption material bottom side, W/K.m^2
h_{tc}	Convection heat transfer coefficient of the top side, W/K.m^2
h_{tr}	Radiation heat transfer coefficient of the top side, W/K.m^2
I(t)	The solar irradiation, W/m^2
k	Thermal conductivity, W/K.m
k_{sorp}	Thermal conductivity of sorption material, W/K.m
q_{equ}	Equilibrium uptake
q_{max}	Sorption material maximum uptake
q_{sorp}	Water uptake
r_p	Sorption material particle radius, m
T_{sorp}	Temperature of sorption material, K
t	Time, s
U_a	Speed of the wind, m/s
V	PV volume, m^3
$\partial q_{sorp}/\partial t$	Water uptake time rate of change
α	Absorptivity of PV layer
β	Tilt angle of PV panel
β_r	Reference temperature coefficient of silicon cell, 1/K
δ	Thickness of PV layer, m
η_{Panel}	Efficiency of Photovoltaic
η_r	Photovoltaic reference efficiency
ε	Emissivity
ε_{glass}	Glass layer emissivity
γ_{air}	Thermal expansion coefficient of air
ν_{air}	Air kinematic viscosity, m^2/s
τ	PV layer Transmissivity

yielding 701.3 mW/m^2 of additional power. Bai et al. [11] further optimized hydrogel evaporative cooling, lowering PV temperatures by 21.9 °C under laboratory conditions while producing freshwater at 0.88 kg/m^2h. Masalha et al. [12] investigated porous media (e.g., sandstone, gravel) integrated into cooling channels. Their multi-method study revealed that optimized porosity (0.35) and flow rates (2 L/min) reduced PV temperatures by up to 35.7 %.

Active methods employ external mechanisms to enhance heat dissipation. Nanofluids, with superior thermal properties, have shown promise. Shaalan et al. [13] reported a 35.7 % temperature reduction using hybrid ZnO/Al_2O_3 nanofluids, improving electrical efficiency by 9.4 %. Satpute et al. [14] designed spiral rectangular absorbers for water cooling, achieving 63.56 % thermal efficiency. However, Sutanto et al. [15] cautioned that nanofluids like Ag-water may reduce transmissivity in critical wavelength ranges, highlighting the need for optical compatibility. Thermal Interface Materials (TIMs) like pyrolytic graphite sheets (PGS) enhance heat transfer. Shurafa et al. [16] demonstrated that PGS reduced PV temperatures by 16 °C, outperforming traditional thermal grease and boosting efficiency by 17.95 %. Abdel-Aziz et al. [17] optimized concentrated PV modules using Ag-water nanofluids, achieving a 55 % reduction in PV surface temperature. However, they noted trade-offs between cooling performance and optical transmissivity, as nanofluids can absorb critical wavelengths of solar radiation, reducing photovoltaic conversion efficiency. Sohail et al. [18] reviewed cutting-edge cooling methods, noting that active techniques (e.g., nanofluids, forced air) achieve higher temperature reductions (up to 39.9 °C) but require energy input.

Hybrid systems combine multiple cooling strategies for synergistic effects. Zhou et al. [19] integrated gas-liquid channels, reducing PV temperatures by 41.65 % and increasing power output by 43.16 %. Fakhraei et al. [20] developed a dual-purpose PVT system with PCMs for passive heating and cooling, improving electrical efficiency by 4.23 % and thermal efficiency by 18.86 %. Li et al. [21] coupled PV modules with humidifier-dehumidifiers (HDHs), achieving a 5.70 % power increase while producing freshwater. Togun et al. [22] explored hybrid PVT systems, emphasizing the role of advanced cooling techniques such as nanofluids and phase change materials (PCMs). Their work highlighted a 15.68 % improvement in thermal efficiency using optimized air-cooled channels,

Optimized designs enhance cooling efficiency. Attia et al. [23] investigated the optimal air-cooling gap width for PVT modules using computational fluid dynamics (CFD). They found that a 5 cm air gap at the top of the PVT module maximized thermal efficiency (25.05 %) and thermal power output (162.31 W) under 1000 W/m^2 solar radiation. Radwan et al. [24] optimized converging channel absorbers using response surface methodology, achieving a PV temperature of 36.6 °C with sandstone porous media. Kabeel et al. [25] introduced concave covers with water and air cooling, improving electrical efficiency by 24.11 %. Attia et al. [26] proposed skeleton-shaped aluminum tubes, boosting thermal efficiency by 24.3 % compared to conventional designs. Srithar et al. [27] designed a bottom-mounted air-cooled PV system with perforated baffles, achieving a 44 % increase in net power output compared to uncooled panels. Abdel-Aziz et al. [28] integrated PV panels with solar water collectors, boosting thermal efficiency by 56.48 % compared to standalone PV systems.

Desiccants such as hygroscopic salts, zeolites, silica gel, metal–organic frameworks (MOF), and hydrogels are examples of materials for evaporative cooling [29–30]. Under ambient circumstances, silica gel is thermally stable, cost-effective, and readily accessible [31]. MOFs' thermal stability varies greatly according on their composition and because of the intricacy of their synthesis and the expense of metal and organic linkers, MOFs are often more expensive to produce [32]. Hygroscopic salts like LiCl and $CaCl_2$ can enhance water absorption when added to hydrogels and silica gel [33]. Silica Gel-$CaCl_2$ sorption composite material showed better performance when compared to Zeolite X13-$CaCl_2$ [34]. Regarding hydrogel, the longevity and functionality of cooling material can be impacted by problems such excessive swelling, adhesion loss, and material deterioration brought on by repeated hydration-dehydration cycles [35].

In another study, Abdel-Aziz and ElBahloul [34] optimized composite sorption materials (Silica Gel-$CaCl_2$) for PV cooling, achieving a 31.11 % temperature reduction at a 3 cm thickness, highlighting the role of material design in thermal management. In a follow-up study, Attia et al. [36] developed a PVT system using bi-fluid cooling (air and CuO-water nanofluids) combined with finned heat exchangers. The system with 48 fins and 1 % CuO nanofluid achieved a thermal efficiency of 78.19 %, representing a 23.57 % improvement over non-finned designs. For MOFs evaporative cooling, Ben-Mansour & El-Maaty [37] used MOF-303 which achieved an improvement in COP of about 0.62 and specific cooling power improved to 970 W/kg. Wang et al. [38] used MOF-801 reduced the temperature of PV panels by up to 14 °C. Liu et al. [39] used MOF-801 which achieved a 25.3 °C reduction in oil temperature compared to natural cooling methods. Wang et al. [40] developed a composite cooling system combining absorbent hydrogel

evaporative (AHE) cooling with a 3D porous copper foam (CF) structure. This design achieved a heat removal rate of 449–713 W/m^2 under 0.8–1.2 sun conditions, reducing PV temperatures to 43–46 °C (vs. 53–66 °C for natural cooling). The CF-AHE system improved electrical efficiency by 4.69–12.84 % in lab tests and 14.01 % in outdoor trials. Yu et al. [41] modeled hydrogel-cooled PV systems, identifying water diffusion within hydrogels as the critical factor for efficiency. Their study showed that increasing hydrogel thickness beyond 6 mm offered diminishing returns, while optimizing diffusion coefficients boosted daily power generation by 2.1 % in Guangzhou, China. This work emphasizes the need to balance heat and mass transfer resistances in hydrogel design. Liu et al. [42] introduced a hygroscopic hydrogel-based passive cooling strategy leveraging nighttime water vapor sorption and daytime evaporation. In lab tests under 1 kW/m^2 irradiance, the hydrogel reduced PV temperatures by 9.9 °C and improved power output by 5.92 %. Outdoor trials in Beijing demonstrated a 7.1 °C temperature drop and 5.21 % power gain, highlighting the method's applicability in diverse climates.

This research pioneers a hybrid passive cooling strategy for PVT systems by integrating Silica Gel-$CaCl_2$ sorption materials with five distinct fin geometries—rectangular, horizontal cylindrical, triangular, trapezoidal, and T-shaped—a novel approach not previously explored in the literature. While prior studies have focused on isolated methods, such as hydrogel evaporative cooling [10,11], PCM stabilization [7,9], or uniform fin designs [5,17,34], this work uniquely bridges sorption thermodynamics with structural innovation. By systematically comparing fin-sorption hybrids, it addresses the critical gap in optimizing both moisture-driven cooling and heat dissipation under high-temperature conditions, offering a dual thermal regulation mechanism absent in earlier single-method solutions. This integration of material science and geometry-specific structural engineering advances PV cooling beyond incremental improvements, enabling tailored designs for arid climates.

The study further distinguishes itself through Multiphysics COMSOL simulations that model coupled heat transfer, fluid dynamics, and sorption-desorption kinetics—an advancement over prior computational analyses limited to isolated parameters [23,24]. Unlike Attia et al. [36], who used fixed fins in bi-fluid systems, or Zhou et al. [19] and Fakhraei et al. [20], who combined active/passive methods like nanofluids with PCMs, this work provides the first comparative framework for fin geometry best performing in sorption-based systems. By correlating geometry with efficiency, durability, and scalability, the findings directly address the material and design challenges noted in earlier research [6,18]. This holistic approach not only refines thermal management but also delivers actionable guidelines for sustainable PV deployment in regions with intense solar radiation, surpassing the limitations of non-adaptive or energy-intensive cooling strategies.

The novelty of this work lies in three aspects: (i) the first systematic integration of sorption material with multiple fin geometries, (ii) the establishment of a comparative framework for geometry-specific comparatively superior in sorption-based cooling, and (iii) the application of a coupled Multiphysics model that simultaneously accounts for heat conduction, sorption–desorption dynamics, and fin-assisted dissipation. These contributions enable actionable guidelines for PV thermal management, particularly in arid and high-irradiance climates such as Cairo, Egypt.

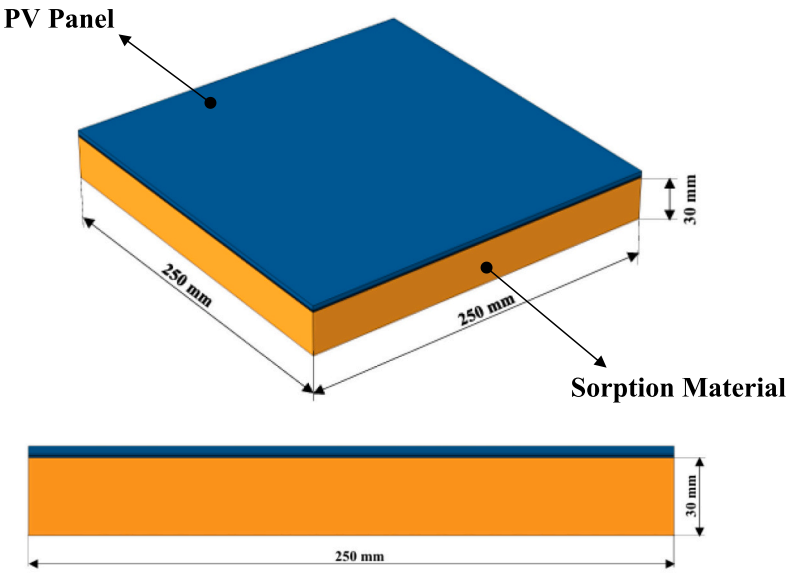
2. Dimensional specifications and scaling factors

This study investigates the integration of an optimized sorption material, specifically a composite of Silica Gel-$CaCl_2$ with a 3 cm thickness, combined with five distinct longitudinal fin geometries positioned below the PV panel and extending through the sorption material. The geometry used in this study was designed to balance realism and computational tractability. The five-layer PV panel consists of glass, ethylene vinyl acetate (EVA), silicon, EVA, and TPT (Tedlar-Polyester-Tedlar) follows thicknesses and material properties consistent with commercial crystalline silicon modules [43]. This ensures that the model reflects realistic heat transfer pathways across the PV laminate. A key innovation in this study is the inclusion of longitudinal fins below the PV panel, which significantly enhance heat dissipation and improve the cooling efficiency of the PV panel [44]. These fins penetrate through the sorption material layer, facilitating effective heat transfer from the PV panel to the sorption material. The fin geometries considered in this study include rectangular, horizontal cylindrical, triangular, trapezoidal, and T-shaped fins. These geometries play a critical role in regulating the thermal behavior of the system by optimizing heat transfer pathways and maximizing surface area for efficient cooling [45]. Also, these fin geometries represent a broad spectrum of conduction-to-surface-area ratios and are commonly reported in cooling applications such as heat exchangers and electronic modules [46–47].

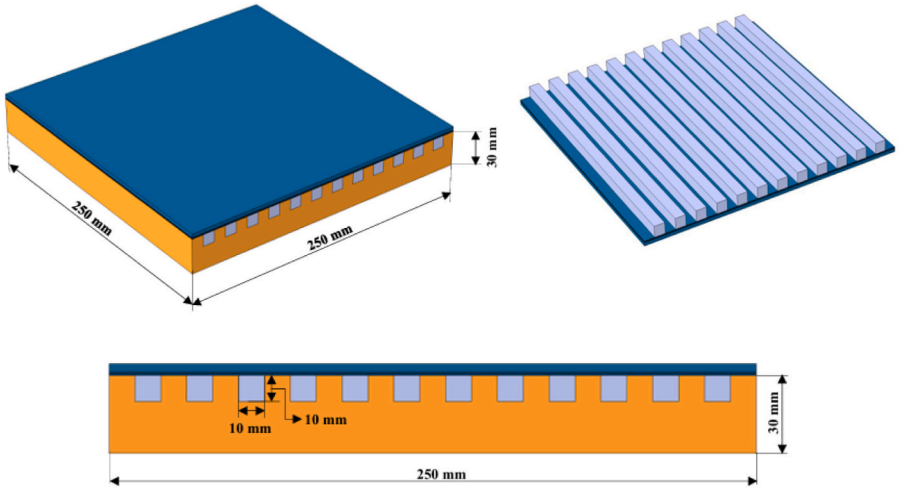
The design of the five fin geometries aims to optimize heat conduction and evaporation efficiency. Rectangular fins offer a large surface area, ensuring steady and uniform heat distribution throughout the sorption layer [48]. Horizontal cylindrical fins promote smooth airflow and reduce thermal resistance, aiding in convective heat transfer [49]. Triangular fins efficiently channel heat toward the sorption material while minimizing material usage, making them a cost-effective option [50]. Trapezoidal fins provide an extended heat transfer surface while maintaining structural integrity for long-term operation [51]. T-shaped fins optimize conduction paths, allowing for a more uniform temperature distribution across the cooling layer [52]. By incorporating these longitudinal fins, the cooling system not only improves heat dissipation but also enhances the evaporation process by ensuring a more uniform temperature distribution within the sorption material [53]. As a result, the cooling efficiency is increased, leading to lower PV temperatures, higher power conversion efficiency, and improved long-term performance.

The proposed PVT system consists of a PV module integrated with longitudinal fins and a composite sorption material layer, as illustrated in Fig. 1. The system runs continually over a 24-h cycle, alternating between two distinct phases: the adsorption phase (night phase) and the desorption phase (day process). During the night phase, as the ambient temperature decreases and the relative humidity increases, the composite sorption material actively adsorbs water vapor from the surrounding air. The adsorption process releases heat, which is subsequently dissipated into the ambient environment, maintaining a relatively lower temperature of the PV panel overnight. The high hygroscopic properties of Silica Gel-$CaCl_2$ enable efficient water vapor adsorption, ensuring that the material is fully saturated before the onset of the daytime cycle. In contrast, during the day phase, solar radiation heats the PV panel, leading to electricity generation but also causing panel temperature to increase. This heat is conducted downward through the longitudinal fins and transferred to the layer of composite sorption material, initiating the desorption process. As water vapor evaporates from the sorption material, it absorbs heat from the system, contributing to the cooling effect of the PV panel. This continuous adsorption-desorption cycle ensures the long-term thermal regulation of the PV panel, preventing excessive heat accumulation and maintaining an optimal operating temperature.

The five-layer PV panel consists of glass, EVA, silicon, EVA, and TPT, as shown in Fig. 2. This multi-layer structure provides mechanical stability, optical transparency, and electrical insulation while ensuring effective solar energy conversion. The composite sorption material layer, placed directly beneath the PV panel and in contact with the longitudinal fins, serves as the thermal management component of the system. The choice of dimensions (3 cm sorption thickness, 10 mm fin depth) was based on prior experimental and theoretical studies [34,36] that demonstrated effective sorption-driven cooling at this scale. The geometry was abstracted to 2D longitudinal cross-sections with fixed length and penetration into the sorption bed to reduce computational time and cost while enabling direct comparison across shapes.



a) PVT system integrated with sorption material



b) PVT system integrated with sorption material and longitudinal rectangular fins

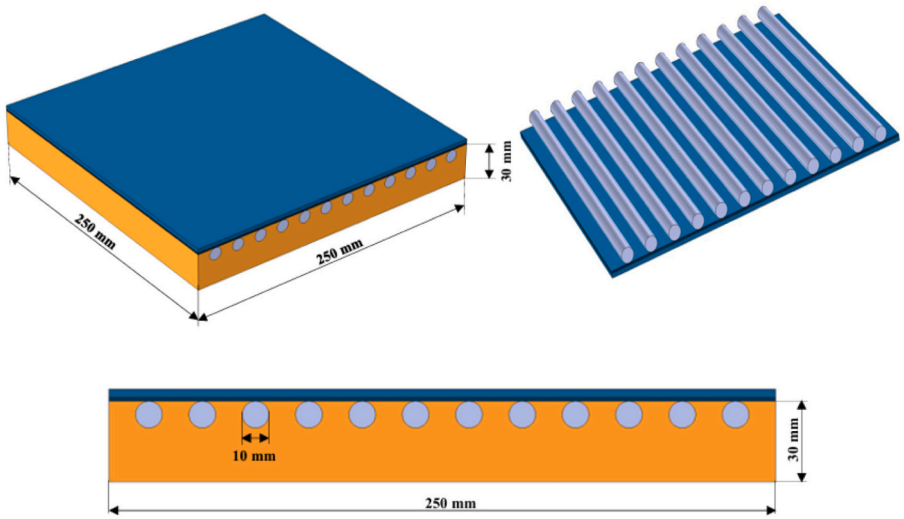
Fig. 1. PVT system geometric configurations.

This research develops a theoretical structure for assessing the operation of a sorption material combined with longitudinal fins for cooling photovoltaic units via Multiphysics simulations. The addition of fins significantly alters heat transfer mechanisms, requiring adjustments to key assumptions [54]. One major simplification is neglecting minor absorption in the above glass layer and side heat losses, as the dominant cooling effects stem from conduction through the sorption material and fins. The fins enhance convective cooling by increasing the heat exchange surface, though variations in heat transfer coefficients due to airflow disruptions are acknowledged. Radiative heat exchange between the PV panel and fins is omitted to focus on conduction and evaporative cooling, with future refinements suggested for wind and radiation modeling.

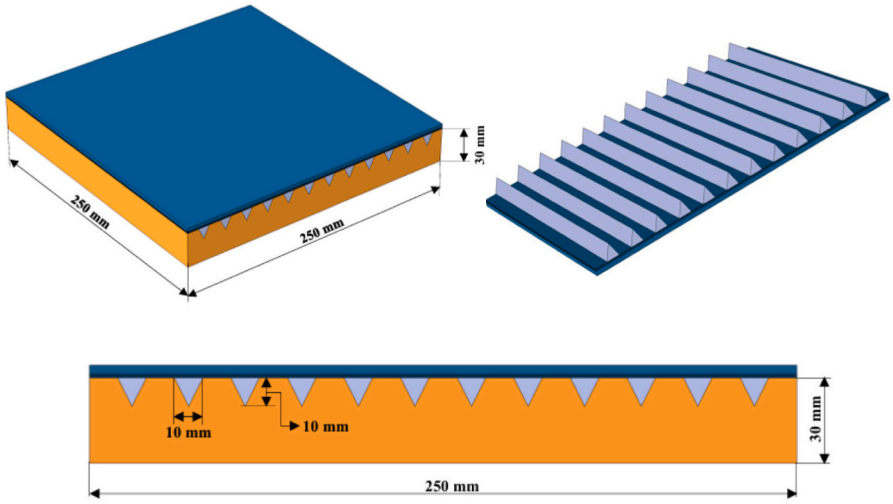
The sorption material is assumed to maintain uniform adsorption/desorption dynamics, though fins may create temperature gradients affecting moisture diffusion. The study assumes initial thermal and humidity equilibrium, but real-world applications may experience transient variations due to the thermal inertia of fins. Contact resistance between the PV panel, sorption layer, and fins is also neglected, though future work should account for potential interfacial thermal resistance. The study solves coupled mass and energy conservation equations, capturing the interaction between heat transfer and moisture evaporation, with considerations for fin heat storage effects. COMSOL Multiphysics software is used for numerical simulations, incorporating heat transfer and convection-diffusion models while refining the fin-sorption interface mesh [55]. These simulations aim to provide insights into best selecting fin-assisted evaporative cooling, guiding future experimental validation and practical implementation in PV cooling applications.

3. Limitations and numerical solution

For PVT systems, thermal regulation relies heavily on three mechanisms: heat conduction, convection, and radiation. The top side surface loses thermal energy via air convection and radiation, even as it absorbs concentrated solar radiation. Simultaneously, convective cooling is the main method of heat dissipation on the bottom side layer. The PVT



c) PVT system integrated with sorption material and longitudinal cylindrical fins



d) PVT system integrated with sorption material and longitudinal triangular fins

Fig. 1. (*continued*).

system initiates at an initial temperature of 293.15 K, featuring adiabatic lower and upper limits to decouple the top and bottom sides heat transfer mechanisms [56]. Effective thermal management across both surfaces is critical to avoid excessive temperature buildup, safeguarding the PVT system's efficiency, longevity, and operational reliability [57].

$$-k\frac{\partial T}{\partial x} = (T_s - T_{glass})h_{tr} + (T_a - T_{glass})h_{tc} \tag{1}$$

where, k is the thermal conductivity in W/K.m, T_s is the sky temperature in K, T_{glass} is glass temperature in K, h_{tr} is the radiation heat transfer coefficient at the top side in W/K.m^2, T_a is ambient temperature in K, and h_{tc} is the convection heat transfer coefficient at the top side in W/K.m^2.

The convection heat transfer coefficient at the top side (h_{tc}) is calculated as [58,59]

$$h_{tc} = 4.07U_a + 5.82 \tag{2}$$

where U_a is the speed of the wind in m/s. With the assumption that the sky is a black body with a temperature of T_s [54]

$$T_s = 0.0552\ T_a^{1.5} \tag{3}$$

The top side radiation heat transfer coefficient, could be obtained using the following formula [54,56]:

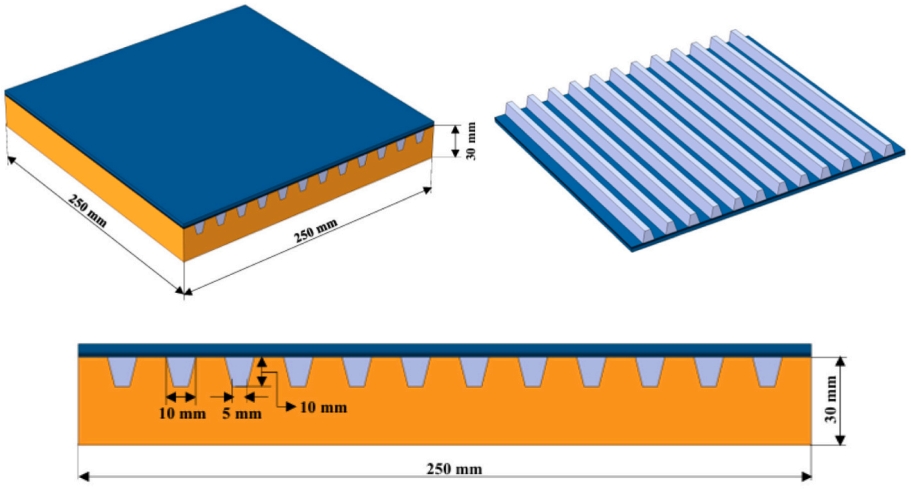
$$h_{tr} = \frac{\left(T_{glass}^4 - T_s^4\right)}{\left(T_{glass} - T_s\right)}\sigma_s\varepsilon_{glass} \tag{4}$$

The bottom surface of the system has the following boundary eq. [57]:

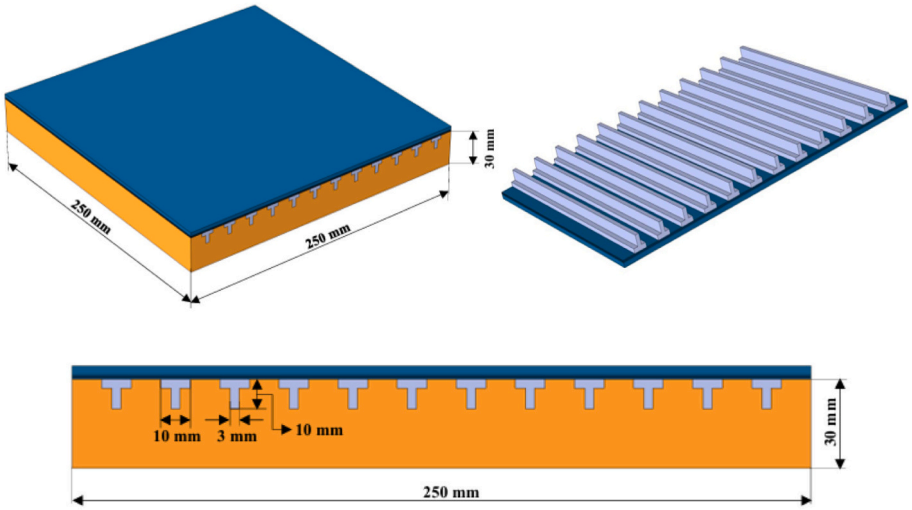
$$-k_{sorp}\frac{\partial T}{\partial x} = h_{bc}\left(T_{amb} - T_{sorp}\right) \tag{5}$$

where, k_{sorp} is the thermal conductivity of sorption material in W/K.m, h_{bc} is the convection heat transfer coefficient of the sorption material bottom side in W/K.m^2, and T_{sorp} is the temperature of sorption material in K. h_{bc} could be computed using the following formula [40]:

$$h_{bc} = \frac{k_{air}Nu}{l_l} \tag{6}$$



e) PVT system integrated with sorption material and longitudinal trapezoidal fins



f) PVT system integrated with sorption material and longitudinal T fins

Fig. 1. (*continued*).

where, k_{air} is thermal conductivity of air in W/K.m, Nu is the Nusselt number, and l_l is characteristic length. Nusselt number could be computed using the following eq. [60]:

$$Nu = \left(0.825 + \frac{0.387 Ra^{1/6}}{\left(1 + \left(\frac{0.492}{Pr}\right)^{9/16}\right)^{8/27}} \right)^2 \tag{7}$$

where Ra is the Rayleigh number and Pr is the Prandtl number, and can be computed from the following eq. [61]:

$$Ra = \frac{g \cos\beta \gamma_{air} \left(T_{sorp} - T_a\right)}{\alpha_{air} \nu_{air}} \tag{8}$$

$$Pr = \frac{\nu_{air}}{\alpha_{air}} \tag{9}$$

where, α_{air} is the air thermal diffusivity in m^2/s, and ν_{air} is the air kinematic viscosity in m^2/s. The thermos-physical characteristics of the air at an average temperature of 40 °C are shown in Table 1. The following relation is used to compute the radiative heat transfer coefficient of the sorption material bottom side [60]:

$$h_{br} = \varepsilon_{sorp} \sigma_s \left(T_s + T_{sorp}\right)\left(T_s^2 + T_{sorp}^2\right) \tag{10}$$

(See Table 2.)

3.1. Photovoltaic panel

The 2D heat transfer model of PV module is evaluated [56,59]

$$\frac{\partial T_j(x,y)}{\partial t} \rho_j C_{p,j} = \left(\frac{\partial^2 T_j}{\partial x^2} + \frac{\partial^2 T_j}{\partial y^2}\right) k_j + Q_j \qquad where\ j = 1, 2, \ldots, n \tag{11}$$

where, $C_{p,j}$ is the specific heat capacity in J/K.kg, k_j is the layer j thermal conductivity in W/K.m, ρ_j is the layer j density in kg/m^3, $T_j(x, y)$ is the layer j temperature, t is the time in sec, and Q_j is the heat source of five

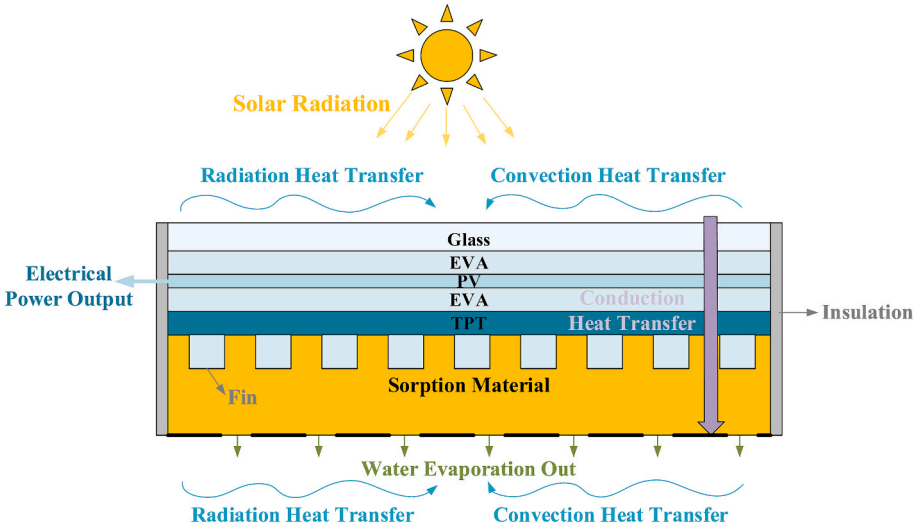


Fig. 2. Heat transfer modes through PVT system combining sorption material with longitudinal fins.

Table 1
Air thermophysical characteristics [61].

Characteristic	Value	Unit
Dynamic viscosity	1.918×10^{-5}	kg/s.m
Specific heat capacity	1007	J/K.kg
Density	1.127	kg/m^3
Thermal conductivity	0.02662	W/K.m
Prandtl number	0.7255	–

Table 2
PV layer characteristics and design specifications [59].

	α	ε	τ	δ [m]	ρ [kg/m^3]	k [W/m.K]	C_p [J/kg.K]
Glass	0.04	0.85	0.92	0.0032	2450	2	500
EVA	0.08		0.90	0.0005	950	0.31	2090
Silicon	0.9		0.02	0.0002	2330	130	677
TPT	0.128	0.9	0.012	0.0003	1200	0.15	1250

layers, as follows [62]:

$$Q_j = \frac{(1-\eta_{Panel})\alpha_j \tau_i I(t) A_j}{V_j} \tag{12}$$

where, $I(t)$ is the solar irradiation in W/m^2, α_j is the absorptivity of the PV layers, τ_i is the prior PV layer transmissivity, A_i is the PV area in m^2, and V_j is the PV volume in m^3. The photovoltaic efficiency η_{Panel} is estimated from [62]:

$$\eta_{Panel} = \eta_r(1-\beta_r(T_{Panel}-T_r)) \tag{13}$$

where, η_r is the photovoltaic reference efficiency which is assumed to be 15 % [63], T_{Panel} is the silicon layer temperature in K, β_r is the reference temperature coefficient of silicon cell (0.0045) 1/K [63], and T_r is the reference temperature which equals to 298 K. Error! Reference source not found. Shows the design characteristics of PV layers.

3.2. Fins material

The study analyses five aluminum fin geometries—rectangular, horizontal cylindrical, triangular, trapezoidal, and T-shaped—with their thermal characteristics summarized in Table 3. These fins are evaluated for their heat dissipation performance, surface-to-volume ratios, and other key thermal properties, leveraging aluminum's advantageous combination of high thermal conductivity, lightweight structure, and corrosion resistance for optimal heat transfer efficiency in photovoltaic systems.

Table 3
Thermophysical Characteristics of aluminum fins [57].

Characteristic	Value	Unit
Specific heat capacity	903	J/K.kg
Density	2675	kg/m^3
Thermal conductivity	211	W/K.m

3.3. Silica gel - $CaCl_2$ composite sorption material

The heat transfer equation for the composite sorption module uses the following relationship [61]:

$$\left(q_{sorp}C_{p,water}m_{sorp} + C_{p,sorp}m_{sorp}\right)\frac{\partial T_{sorp}}{\partial T} = -Ah_{bc}\left(T_{sorp}-T_a\right) - Ah_{br}\left(T_{sorp}-T_s\right) + \frac{\partial q_{sorp}}{\partial t}m_{sorp}H_s \tag{14}$$

where, h_{br} is the radiation heat transfer coefficient for the sorption material bottom side in W/K.m^2, h_{bc} is the convection heat transfer coefficient of the sorption material bottom side in W/K.m^2, $C_{p,water}$ is the water specific heat capacity in J/K.kg, m_{sorp} is the mass of sorption material in kg, $C_{p,sorp}$ is the sorption material specific heat capacity of in J/K.kg, q_{sorp} is water uptake, T_{sorp} is the temperature of sorption material in K, H_s is the isosteric heat of adsorption of the sorption material in kJ/kg, and $\partial q_{sorp}/\partial t$ is the water uptake time rate of change. The equilibrium water uptake rate, $\partial q_{sorp}/\partial t$:

$$\frac{\partial q_{sorp}}{\partial t} = 15\frac{D_{sd}}{r_p^2}\left(q_{equ}-q_{sorp}\right) \tag{15}$$

where, r_p sorption material particle radius in m, q_{equ} is the equilibrium water uptake, which could be computed using the following formula:

$$q_{equ} = q_{max}exp\left(-\left(\frac{RuT_{sorp}}{E_c}ln\left(\frac{p_{sat}(T_{sorp})}{p_{vap}}\right)\right)^n\right) \tag{16}$$

where E_c is the characteristic energy in J/mol, q_{max} is the sorption material maximum uptake, and Ru is the universal gas constant. The saturation pressure of water $p_{sat}(T)$ at a given temperature T, is

computed by using the following eq. [64]:

$$p_{sat}(T) = 8.143 \times 10^{10} exp\left(\frac{-5071.7}{T}\right) \quad (17)$$

The atmospheric water vapor pressure is obtained using the following relationship [64]

$$p_{vap} = RH \times p_{sat}(T_a) \quad (18)$$

The surface diffusivity D_{sd} of water within the sorption material layer is computed from [64],

$$D_{sd} = D_{sdo} exp\left(-\frac{E_{act}}{RuT_{sorp}}\right) \quad (19)$$

where, D_{sdo} refers to the surface diffusivity pre-constant in m^2/s and E_{act} is the activation energy in J/mol. This present model simulates the composite sorption material for Silica gel-$CaCl_2$ and the parameters of the Silica gel-$CaCl_2$ is presented in Table 4.

3.4. Boundary conditions

The upper surface of the PV module was modeled as exposed to solar irradiation I(t), with heat losses governed by convection to the ambient air and radiation to the sky (Eqs. (1) to (4)). The sky was treated as a black body with effective temperature T_s, while the ambient air temperature followed the diurnal weather profile of Cairo. The bottom surface of the sorption layer exchanged heat by natural convection and radiation (Eqs. (5) to (10)), with ambient reference pressure set to 1 atm. The lateral boundaries of the computational domain were treated as adiabatic, given the symmetry and the focus on vertical heat transfer.

A mesh sensitivity study was conducted using five mesh densities: 0.00925 m, 0.0025 m, 0.002 m, 0.001 m, and 0.0009 m. The PV temperature profile was monitored over 24 h for each mesh configuration. As shown in Fig. 3, negligible differences (<0.2 °C) were observed between the two finest meshes (0.001 m and 0.0009 m), confirming numerical convergence. The 0.0009 m mesh was adopted as the baseline for all subsequent simulations, as it offered the best balance between accuracy and computational cost.

Mesh independence is a critical concept in CFD that ensures the numerical results of a simulation are not significantly affected by the resolution or size of the computational mesh. When a simulation achieves mesh independence, further refinement of the mesh does not lead to substantial changes in the results, indicating that the solution has converged to a reliable approximation of the physical problem. This is essential for obtaining accurate and trustworthy predictions, as overly coarse meshes may miss important flow features, while excessively fine meshes can be computationally expensive without providing additional meaningful information. To determine mesh independence, a mesh sensitivity study is conducted by running simulations with progressively finer meshes and comparing key output parameters (e.g., temperature, velocity, or pressure). In the provided study, a structured quadrilateral mesh was used in the 2D model, refined near the PV–fin–sorption interfaces to capture steep temperature and moisture gradients. Special

Table 4
Properties of Silica gel - $CaCl_2$l [64].

Parameter	Value	Unit
Radius, r_p	1.7×10^{-4}	m
Heterogeneity parameter, n	1.15	
Maximum uptake, q_{max}	0.8	
Pre-exponential diffusion parameter, D_{sdo}	2.54×10^{-4}	m^2/s
Thermal conductivity, k_{sorp}	0.198	W/K.m
Density, ρ_{sorp}	740	kg/m^3
Specific heat, $C_{p,sorp}$	924	J/K.kg
Isosteric heat of adsorption, H_s	2760	kJ/kg
Activation energy, E_{act}	42,000	J/mol
Porosity	0.37	

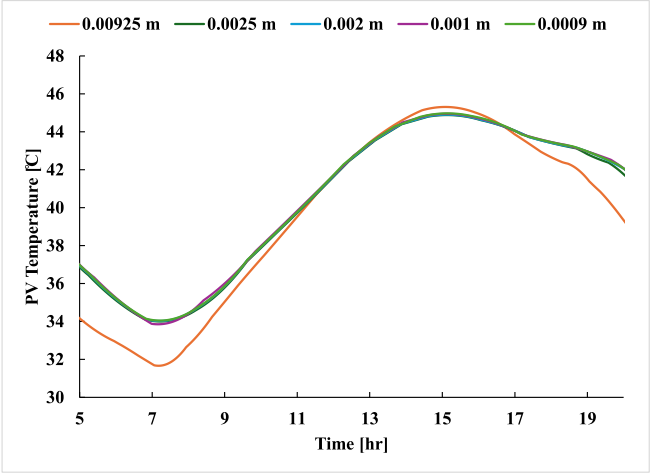
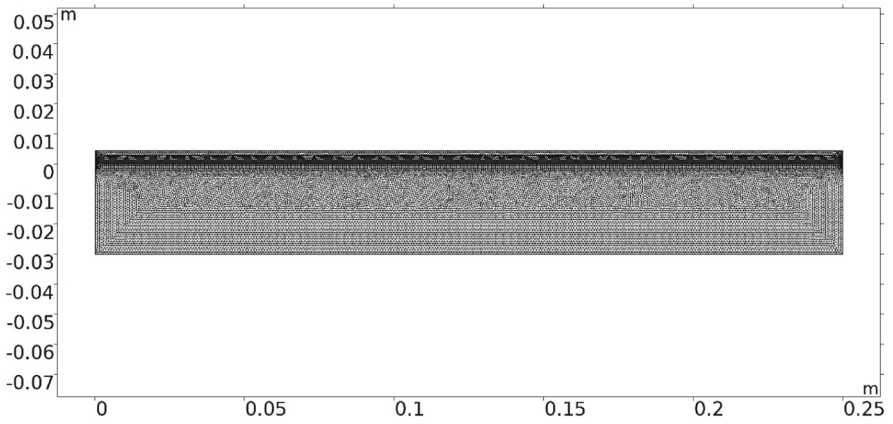


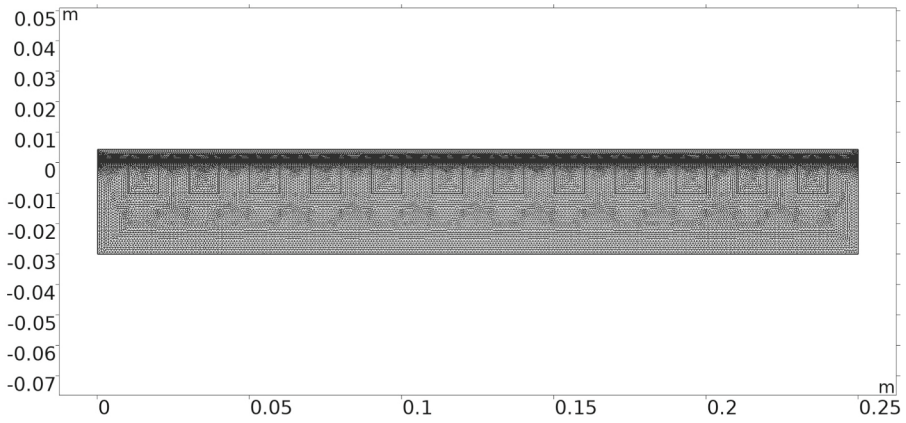
Fig. 3. PV temperature versus time for different mesh size.

care was taken to resolve the narrow conduction pathways along fin geometries and the sorption layer thickness (3 cm). Mesh independence was tested by comparing results across five different mesh sizes: 0.00925 m, 0.0025 m, 0.002 m, 0.001 m, and 0.0009 m, see Fig. 3. The results showed that the differences between the 0.001 m and 0.0009 m meshes were negligible, suggesting that the solution had converged. The minimum mesh element size near the fin tip was 9×10^{-4} m, while coarser elements ($\approx 3 \times 10^{-3}$ m) were applied in regions with low gradients, ensuring computational efficiency. The 0.0009 m mesh was selected as it provided a balance between computational efficiency and accuracy, capturing localized temperature gradients and heat transfer phenomena (conduction, convection, and radiation) with sufficient precision. In the context of thermal analysis for PVT systems, mesh independence is particularly important because it ensures that localized temperature variations, which can significantly impact PV performance and durability, are accurately resolved. For example, overheating risks and thermal management efficiency depend on capturing small-scale temperature gradients that may arise from geometric features like fins or cooling channels. The selected mesh size (0.0009 m) aligns with CFD best practices, as it reliably predicts these gradients without unnecessary computational overhead.

The data shown in Fig. 3 further supports the mesh independence claim. For instance, the temperature profiles over time for the finest meshes (0.001 m and 0.0009 m) are nearly identical, with only minor deviations. This consistency confirms that the chosen mesh resolution is adequate for the study's objectives. Achieving mesh independence not only validates the simulation setup but also enhances confidence in the results, ensuring that conclusions about PV performance and thermal management are based on robust numerical foundations. The time-dependent test was conducted with measurements recorded at intervals of 1 min, 5 min, and 10 min. The three resulting curves were almost identical, showing no observable differences. This confirms that the model is stable and its thermal response remains consistent regardless of the chosen time step. All simulations were performed using COMSOL Multiphysics under time dependent solver. To ensure the accuracy of the numerical results, convergence criteria were clearly defined and monitored during the simulations. A relative tolerance of 10^{-4} and an absolute tolerance of 10^{-6} were applied in the COMSOL Multiphysics solver. Convergence was confirmed when the residuals of all governing equations dropped below these thresholds. Fig. 4 shows the two-dimensional mesh representation of a PVT system with evaporative cooling combined with different fin geometrics.



a) PVT evaporative cooling system without fin



b) PVT evaporative cooling system with longitudinal rectangular fin

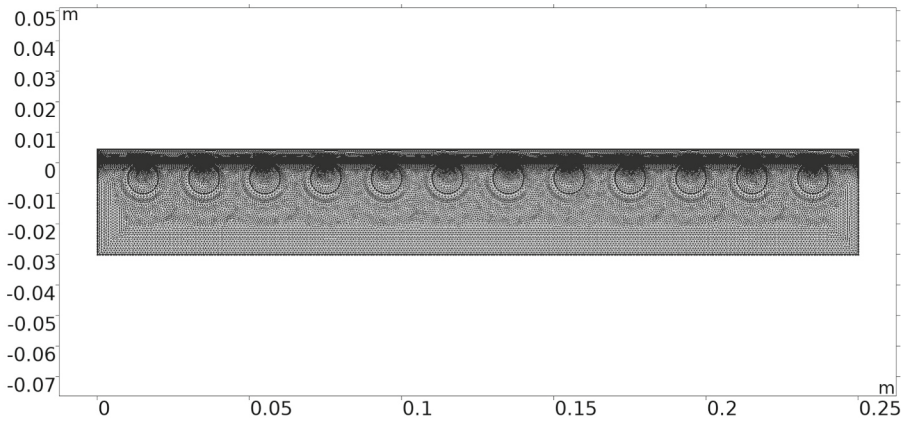
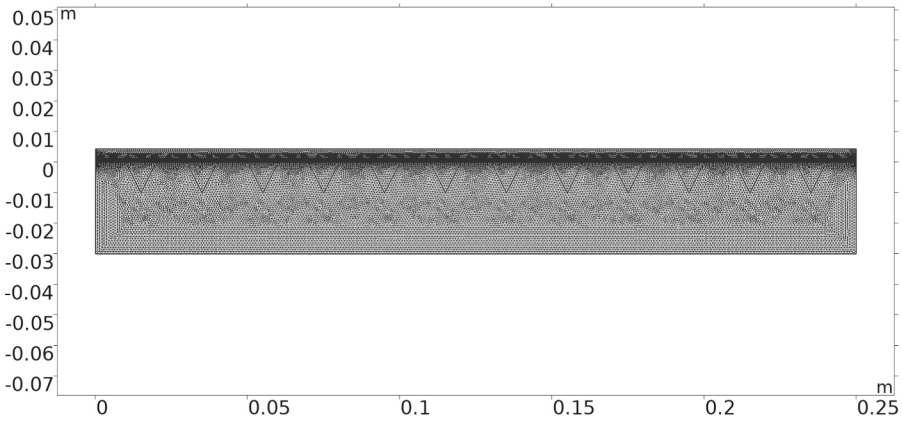
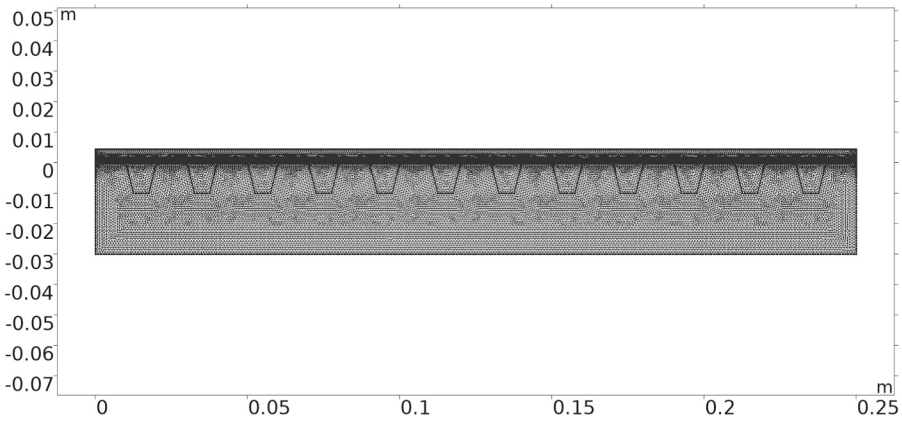


Fig. 4. A 2 D uniform mesh representation of a PVT evaporative cooling system with different longitudinal fins.

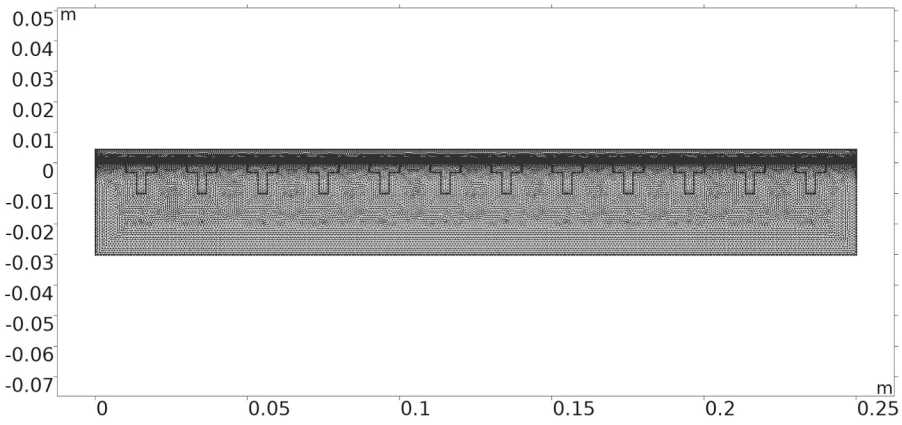
c) PVT evaporative cooling system with longitudinal cylindrical fin



d) PVT evaporative cooling system with longitudinal triangle fin



e) PVT evaporative cooling system with longitudinal horizontal fin



f) PVT evaporative cooling system with longitudinal T shaped fin

Fig. 4. (*continued*).

4. Model validation

The numerical model of the PVT evaporative cooling system is validated in two validation steps. First step is carried out by comparing the CFD predicted PV panel temperatures model finding against the published experimental measured data presented by Radwan et al. [24]. PV temperature comparison between present CFD model and Radwan et al. [24] experimental results is shown in Fig. 5. The numerical results exhibit significant agreement with a relatively average error of 2.25 % compared with the experimental data of Radwan et al. [24]. This close correlation between the present model and Radwan et al. [24] results confirm the accuracy and reliability of the developed numerical model for simulating PV temperature behavior under different environmental conditions. The consistent trend and minimal deviation across the tested ranges demonstrate that the model effectively captures the thermal response of the PV panel, thereby establishing a solid foundation for further investigations into the performance of the PVT evaporative cooling system under more complex configurations.

Second step is carried out by comparing the CFD predicted sorption

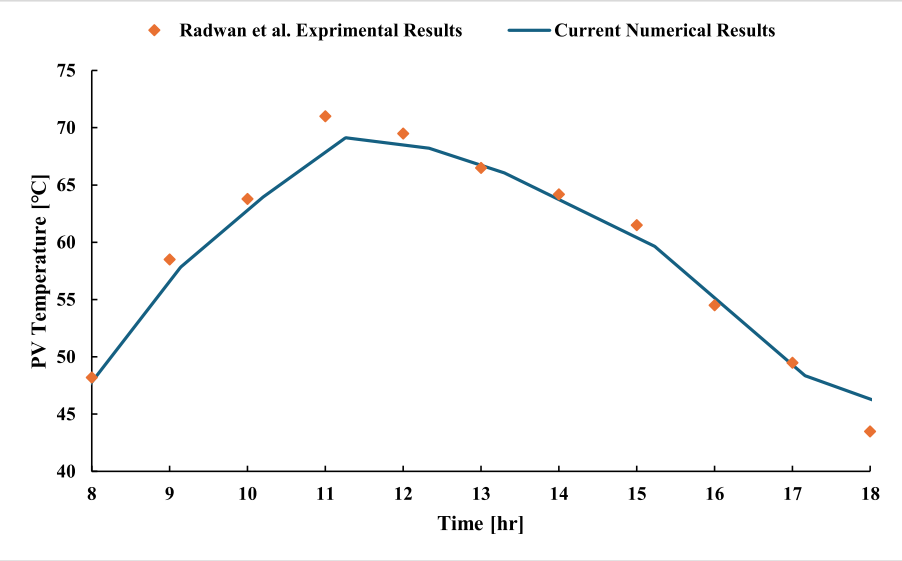


Fig. 5. PV temperature comparison between present CFD model and Radwan et al. [24] experimental results.

model finding with the published experimental measured data resulted from Mohammed et al. [63] which ensures the model's accuracy and reliability in predicting uptake and sorption bed temperature behavior. The comparison, illustrated in Fig. 6(a) and Fig. 6(b), shows that the model captures the average uptake and sorption bed temperature trends observed in the experiments, confirming its ability to evaluate relative performance improvements between different cooling configurations. The numerical results exhibit significant agreement with a relatively average error of 2.6 % for uptake and 4.6 % for sorption bed temperature compared with the experimental data of Mohammed et al. [63]. This alignment between simulated and experimental results demonstrates the model's robustness in replicating real-world behavior. The validation process highlights the model's strengths in capturing key dynamics while also identifying areas for refinement. The overall agreement between the model and experimental data supports its use as a predictive tool for system performance. The maintained thermal trends indicate that the model reliably reflects the underlying physics, making it suitable for comparatively superior design and comparative analysis. This validation step is essential for ensuring confidence in the model's outputs and its applicability to practical engineering scenarios.

4.1. Uncertainty and sensitivity analysis

To quantify robustness, a brief sensitivity analysis was conducted by perturbing three critical input parameters: solar irradiance (±5 %), wind speed (±5 %), and ambient temperature (±5 %). Results indicated limited influence on the overall performance trends. The largest variation was observed in peak PV temperature, which changed by a maximum of **±3.5 %**, while average efficiency varied by less than **±2.1 %**. Importantly, the relative ranking of fin geometries remained unchanged, confirming that the comparative conclusions are insensitive to moderate uncertainty in boundary conditions or material properties.

5. Results and discussions

The study utilizes real-world climate data from Cairo City (30.02°N, 31.13°E) on July 12, 2024, to simulate realistic operating conditions for the PV cooling system. The 24-h dataset captures key environmental variables: ambient temperature (T_a) peaks at 43.58 °C at 14:00, solar irradiance reaches 976.33 Wh/m^2 at 11:00, and wind speed varies from 1.71 m/s (11:00) to 8.5 m/s (20:00), see Fig. 7. These parameters critically influence the PV system's thermal behavior—higher irradiance increases heat generation, while wind speed enhances convective cooling. The absence of solar radiation at night (0:00–5:00 and 19:00–23:00) allows analysis of passive cooling effects, particularly the sorption material's role in heat dissipation during non-operational hours. The selected weather profile represents a typical summer day in Cairo, characterized by intense solar exposure and significant diurnal temperature fluctuations. The data shows that peak irradiance (~976 Wh/m^2) precedes the highest ambient temperature (43.58 °C) by 3 h, reflecting heat accumulation in the system. Wind speeds are lowest during peak sunlight (11:00–12:00), reducing convective cooling when heat generation is maximal, and rise sharply in the evening (peaking at 8.5 m/s), accelerating nighttime cooling. This dynamic interplay between irradiance, temperature, and wind speed provides a rigorous test for evaluating the hybrid cooling system's performance under realistic stress conditions, ensuring the findings are applicable to real-world PV deployments in hot climates.

5.1. PV surface temperature

Fig. 8 compares the temperature profiles of PVT systems under seven different cooling configurations over a 24-h period in Cairo, Egypt. The uncooled PV module reaches peak temperatures of around 66.9 °C at midday (12:00), while the integration of a sorption material (Silica Gel-$CaCl_2$) alone reduces this peak to 56 °C. The most effective cooling is achieved by combining the sorption material with fin geometries, particularly rectangular fins, which lower the peak temperature to 38.6 °C with a reduction of nearly 28.3 °C compared to the uncooled module. The performance of the fins follows a clear hierarchy: rectangular fins outperform horizontal cylindrical, triangular, trapezoidal, and T-shaped fins, highlighting the importance of fin design in maximizing heat dissipation. The cooling effectiveness varies significantly with solar irradiance, showing the greatest impact during peak sunlight hours (10:00–14:00). At night, when irradiance is negligible, all configurations converge to similar temperatures, indicating that the cooling mechanisms are primarily active during daylight. The sharp rise in PV temperature between 10:00–14:00 corresponds to the period of maximum irradiance and lowest wind speed, when heat accumulation in the silicon layer is greatest. Hybrid fin–sorption systems show markedly lower temperature rise, with rectangular fins achieving a 28.3 °C reduction compared to the uncooled case. This trend is consistent with fin theory, as rectangular fins maximize surface area for conduction and enhance sorption-driven cooling.

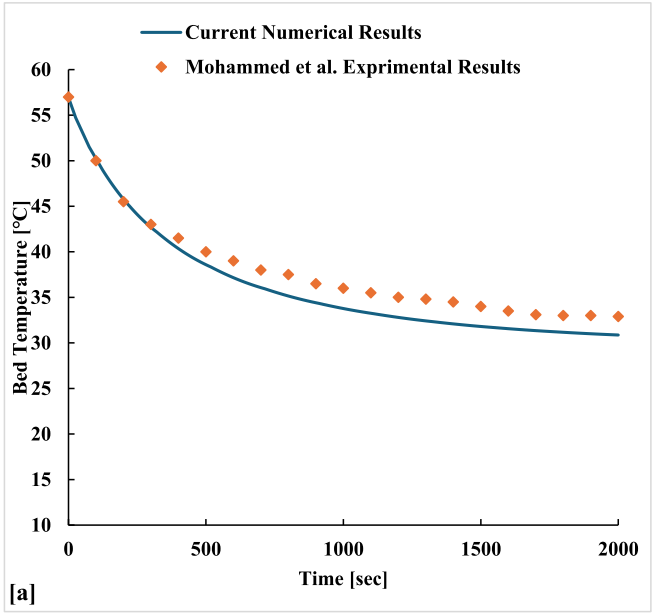


Fig. 6. Comparison of Mohammed et al. [63] experimental results and current numerical results: (a) uptake profile, (b) temperature profile.

The thermal trends observed in this study could be explained by the interplay of conduction, convection, radiation, and sorption–desorption dynamics. For conduction dynamics, heat generated in the silicon layer was primarily conducted downward through the PV stack into the fins and sorption bed. Fin theory showed that rectangular fins, with their high surface-to-volume ratio and uniform cross-section, achieve the highest fin efficiency, distributing heat evenly into the sorption layer. Cylindrical fins, by contrast, have a smaller wetted perimeter and lower conduction effectiveness, which explains their weaker cooling performance. For convective and buoyancy effects, the system lost heat from the top surface by forced convection with wind and from the bottom by natural convection governed by the Rayleigh and Nusselt relations. During midday, when wind speeds were low and irradiance was high, the fin–sorption pathway dominated heat dissipation. In the evening, higher wind speeds increased convective heat transfer and accelerate cooling, explaining the convergence of temperatures across all configurations. For radiative exchange, long-wave radiation to the sky contributed significantly at night, reducing overall system temperature. This explains why differences between configurations diminish during non-irradiated hours.

Fig. 9 compares the average operating temperatures of photovoltaic (PV) modules under seven different cooling configurations. The uncooled PV system maintains the highest average temperature at 54.01 °C, demonstrating the significant thermal stress experienced in hot climates like Cairo. The introduction of sorption material (Silica Gel-$CaCl_2$) alone reduces this by 11.12 °C to 42.89 °C, while hybrid systems combining sorption with fin geometries achieve even better cooling. Among these, rectangular fins perform best with an average temperature of just 37.75 °C - a remarkable 16.26 °C reduction from the baseline - followed closely by triangular (38.06 °C) and T-shaped fins (38.04 °C), while cylindrical fins show the least improvement (39.77 °C). The average temperature hierarchy (rectangular < triangular ≈ T-shaped < trapezoidal < cylindrical < uncooled) reflects the balance between fin efficiency and sorption interaction. Cylindrical fins, with lower perimeter-to-area ratios, exhibit reduced effectiveness, confirming the importance of geometry selection for thermal design. These findings highlight that while sorption materials provide substantial passive cooling benefits, their combination with properly designed fins especially rectangular ones creates the most effective thermal management solution for PV systems in high-temperature environments.

For validation against experimental data (Mohammed et al. [63]), reported uncertainties in PV temperatures were included as error bars in Fig. 9. This representation highlights that the model consistently falls within experimental variability bounds, further supporting its reliability.

5.2. PVT system efficiency

Fig. 10 presents the PVT efficiency over time for seven different configurations: an uncooled PV module and six cooled variants, including one with only sorption material and five with sorption material combined with different fin geometries (rectangular, cylindrical, triangular, trapezoidal, and T-shaped). The data shows that the uncooled PV module exhibits the lowest efficiency, starting at around 14.64 % and gradually declining over time due to rising temperatures. In contrast, all cooled configurations maintain higher efficiencies, with the rectangular fin design performing the best, retaining the highest efficiency values throughout the 12-h period. This underscores the effectiveness of active cooling in mitigating temperature-related efficiency losses. This highlights how fin geometry influences cooling performance. For instance, the rectangular fin configuration consistently outperforms the others, maintaining efficiency levels above 14.3 % even after 12 h, while the uncooled module drops below 13.34 %. The sorption material alone improves efficiency compared to the uncooled case, but the addition of fins further enhances heat dissipation, with the rectangular fin offering the largest surface area for heat transfer. These results validate the computational model's predictions and demonstrate that comparatively superior fin designs, particularly rectangular fins, can significantly boost PV efficiency by maintaining lower operating temperatures. Fig. 11 compares the hourly PVT efficiency across seven different cooling configurations, demonstrating how passive and hybrid cooling methods improve performance compared to an uncooled PV module. The uncooled PV system has the lowest average efficiency (13.03 %), as it suffers from significant heat buildup, which reduces electrical output. Introducing sorption material (Silica Gel-$CaCl_2$) alone improves efficiency to 13.78 %, confirming that moisture absorption and evaporative cooling help lower panel temperatures. Further enhancements are achieved by combining sorption material with different fin geometries, which increase heat dissipation. The rectangular fin performs best, achieving an average efficiency of 14.13 %, followed closely by the T-fin and triangular fin (14.11 %) and the trapezoidal fin (14.10 %). The cylindrical fin shows a slightly lower improvement (13.99 %), likely due to its reduced surface area compared to flat or

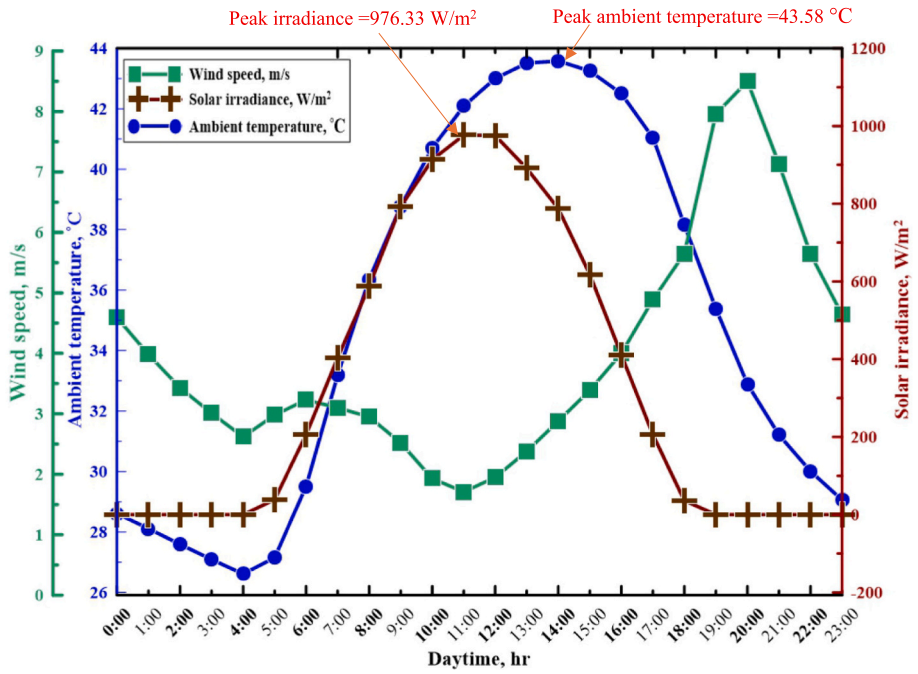


Fig. 7. Weather conditions at Cairo city (30.02°N, 31.13°E) on July 12, 2024 used as boundary inputs for simulations.

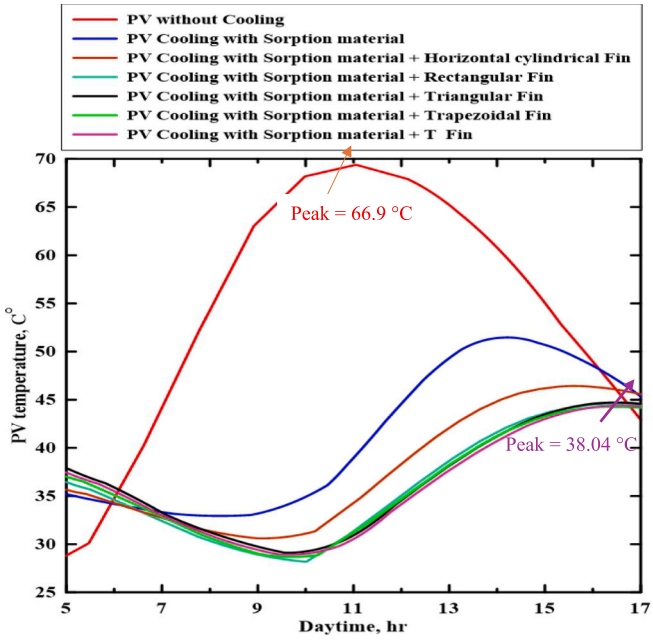


Fig. 8. Hourly PV surface temperature under different cooling configurations.

structured fins. Panel efficiency declines over time in the uncooled case due to rising temperature, dropping below 13.34 %. In contrast, rectangular fins maintain efficiencies above 14.1 % throughout the operating period. The results demonstrate that comparatively superior fin geometries not only lower peak temperatures but also stabilize efficiency under thermal stress. These results highlight that fin geometry plays a crucial role in cooling performance, with flat and extended-surface designs (rectangular, T-fin, triangular) being more effective than curved ones (cylindrical). Overall, the combination of sorption material and best performing fin structures maximizes PVT efficiency by maintaining lower operating temperatures.

5.3. PVT system uptake

Fig. 12 presents the uptake values of the sorption material (Silica Gel-CaCl2) over time for different fin geometries integrated with a PVT cooling system. The uptake values, which indicate the material's ability to absorb moisture or heat, generally increase over time but slow down as saturation approaches. Fig. 12 compares six configurations: a PVT module cooled by sorption material alone, and modules cooled by sorption material with rectangular, cylindrical, triangular, trapezoidal, and T-shaped fins. The results show that finned configurations have higher uptake values than the sorption-only case, suggesting that fins enhance heat dissipation and improve cooling performance. Among the tested geometries, rectangular fins demonstrate the highest uptake values at many time points, indicating superior cooling efficiency. Cylindrical and trapezoidal fins also show good performance. The uptake values stabilize after approximately 17 p.m., suggesting that the sorption material reaches its capacity, making periodic regeneration necessary for sustained efficiency. The data highlights that integrating comparatively superior fin geometries with sorption-based cooling significantly improves PV module cooling, helping maintain efficiency in high-temperature environments. Sorption–desorption coupling dynamics showed that the latent sink term associated with desorption provided strong cooling during peak solar hours. Fins enhanced this process by conducting heat efficiently into the sorption bed, raising local vapor pressure gradients and sustaining stronger desorption. The result is higher uptake values and improved cooling in finned systems, particularly with rectangular geometries. Rectangular fins achieved the most uniform heat spreading and the largest uptake (0.346), resulting in the lowest peak temperature (38.6 °C). Triangular and T-shaped fins provide efficient conduction paths with moderate material usage, while trapezoidal fins balance conduction and structural integrity. Cylindrical fins show the weakest performance due to their reduced perimeter-to-area ratio. Interplay of conduction, convection, radiation, and sorption–desorption dynamics clarify the observed hierarchy of performance (rectangular > triangular ≈ T-shaped > trapezoidal > cylindrical) and highlight how coupled thermal-fluid and sorption phenomena govern PVT system behavior under real-world conditions.

Fig. 13 presents the average uptake values of the sorption material (Silica Gel-$CaCl_2$) for different PVT cooling configurations. The data

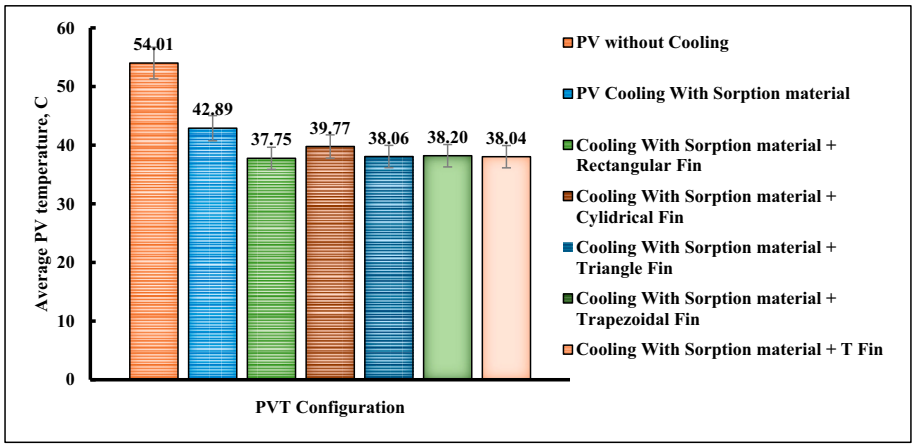


Fig. 9. Average PVT operating temperature across configurations.

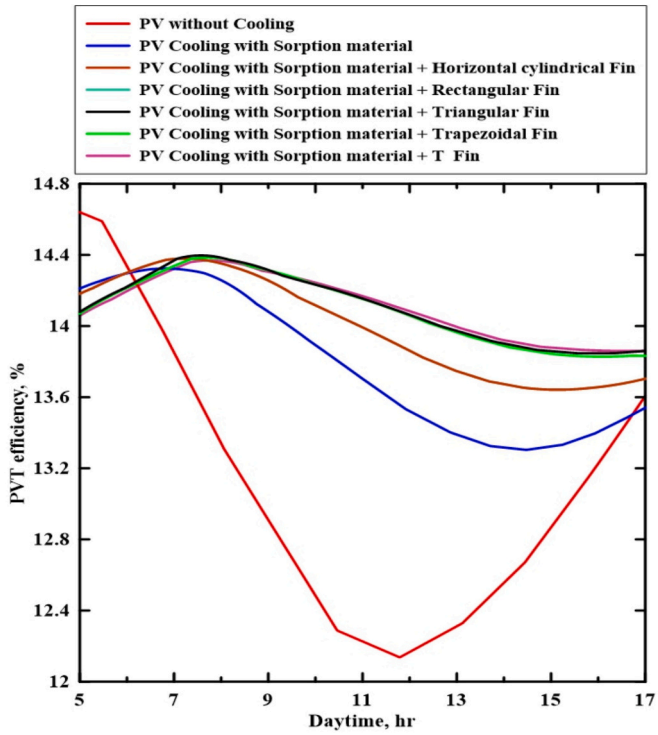


Fig. 10. Hourly PVT efficiency over a 12-h cycle.

shows that the baseline case, where the PV module is cooled only with the sorption material, has the lowest average uptake value (0.302). This indicates less moisture or heat absorption due to limited heat dissipation. When fins are integrated, the uptake values increase, suggesting enhanced heat transfer efficiency. Among the fin configurations, the rectangular fin has the highest average uptake (0.346), followed closely by the triangular (0.341), T-shaped (0.341), and trapezoidal (0.340) fins, while the cylindrical fin records a slightly lower uptake (0.330). The higher uptake values in finned configurations suggest that the added surface area and improved heat dissipation enable the sorption material to absorb more moisture or heat. Rectangular fins show the highest uptake, possibly due to their extended surface contact, while cylindrical fins perform slightly less effectively. The similarity in uptake values among triangular, trapezoidal, and T-shaped fins indicates comparable cooling performance. The sorption-only configuration shows limited uptake (0.302) due to slower heat transfer. Fin integration increases uptake by enhancing local desorption rates, with rectangular fins reaching 0.346. The convergence of uptake values after 17:00 indicates saturation of the sorption material and alignment with the evening cooling phase. This data reinforces that integrating best performing fin geometries with sorption materials enhances PV cooling efficiency, contributing to better thermal regulation and overall system performance.

6. Practical applications & limitations

For practical engineering implications, the simulation outcomes demonstrate that integrating sorption material with comparatively

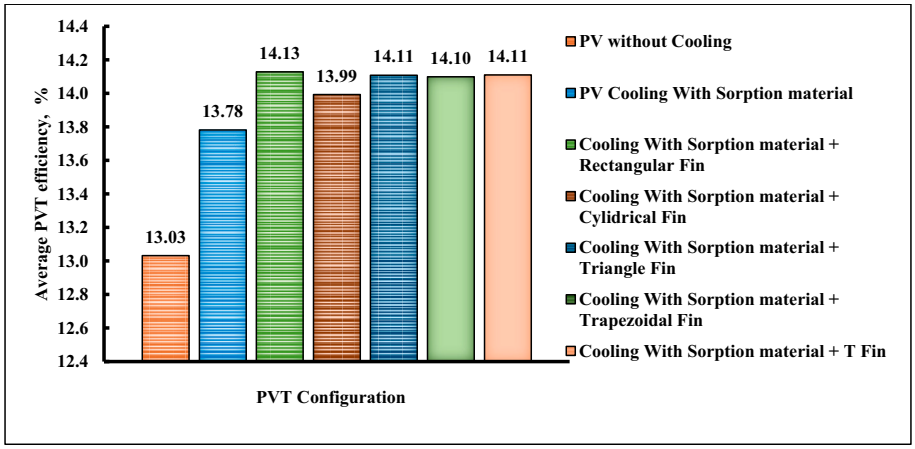


Fig. 11. Average PVT efficiency for all configurations.

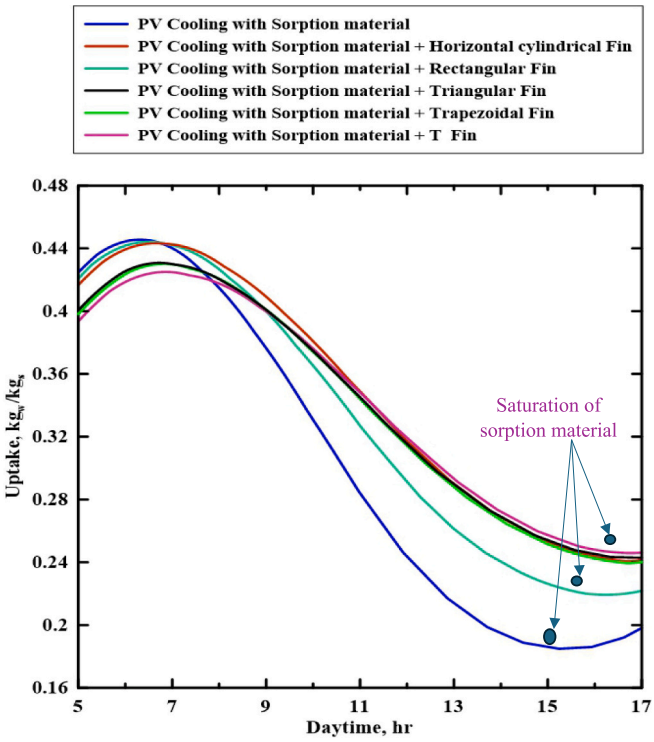


Fig. 12. Hourly uptake values of sorption material (Silica Gel–$CaCl_2$) for different configurations.

superior fin geometries can reduce PV operating temperature by up to 28.3 °C and improve electrical efficiency by 8.4 % compared to uncooled modules. These improvements are significant in real-world contexts where PV modules are often deployed in hot-arid climates such as North Africa, the Middle East, and southern Asia, where peak daytime cell temperatures frequently exceed 65 °C. By reducing overheating, the proposed system can extend PV module lifespan, improve daily energy yield, and lower the levelized cost of electricity (LCOE). For large-scale PV farms, even a 1 % efficiency gain translates into substantial energy savings over the system lifetime, making the ~1.1 percentage-point gain observed here highly valuable. One of implementation considerations that the design leverages passive processes (sorption–desorption cycles and fin conduction), which eliminates the need for external pumps, fans, or energy input, reducing operational costs and maintenance complexity. The rectangular fin configuration, identified as the most effective, is also the most straightforward to manufacture using aluminum extrusion or sheet fabrication, suggesting feasibility for scaling. Given its simplicity and passive nature, the proposed system could be particularly valuable for utility-scale solar farms in hot deserts (e.g., Middle East), building-integrated PV systems, where passive cooling reduces building heat load and off-grid solar applications in rural or remote areas, where maintenance-free solutions are critical. These practical insights confirm that the proposed hybrid cooling design is not only computationally effective but also technically transferable to real-world solar energy systems, provided that durability and material regeneration are further studied. Despite these benefits, several factors should be addressed before large-scale implementation:

- **Material regeneration:** Sorption materials (Silica Gel–$CaCl_2$) gradually saturate and require nighttime adsorption or periodic regeneration cycles. In high-humidity regions, desorption capacity may decline over time.
- **Durability:** Long-term stability of the composite sorption layer under cyclic heating–cooling and moisture exchange requires experimental validation.
- **Structural integration:** Embedding fins within PV back sheets may require design modifications to preserve electrical insulation and mechanical integrity.
- **Climatic dependence:** While effective in hot-arid climates, the system's relative advantage may diminish in cooler or windy climates where natural convection dominates.

Although the numerical results identified rectangular fins as the best-performing geometry in terms of maximum heat uptake and peak PV temperature reduction, practical considerations must also be evaluated before large-scale application:

- **Manufacturing feasibility:** Rectangular and cylindrical fins are the simplest to manufacture using conventional extrusion or sheet-metal stamping. In contrast, T-shaped and trapezoidal fins require more complex machining or casting, which may increase production cost.
- **Compactness and integration:** Rectangular fins offer straightforward integration with sorption layers but add bulk in terms of depth. Cylindrical fins are more compact but offer lower surface-to-volume ratios. Triangular and trapezoidal fins provide a compromise between compactness and thermal performance.
- **Weight and cost considerations:** While rectangular fins maximize cooling, they require more material per unit volume, leading to higher weight and cost. Lightweight triangular fins, though slightly less efficient, may be more practical for portable or modular PV applications.
- **Sorption duration and thermal cycling:** Fins with larger contact areas (rectangular and T-shaped) enhance initial sorption rates but

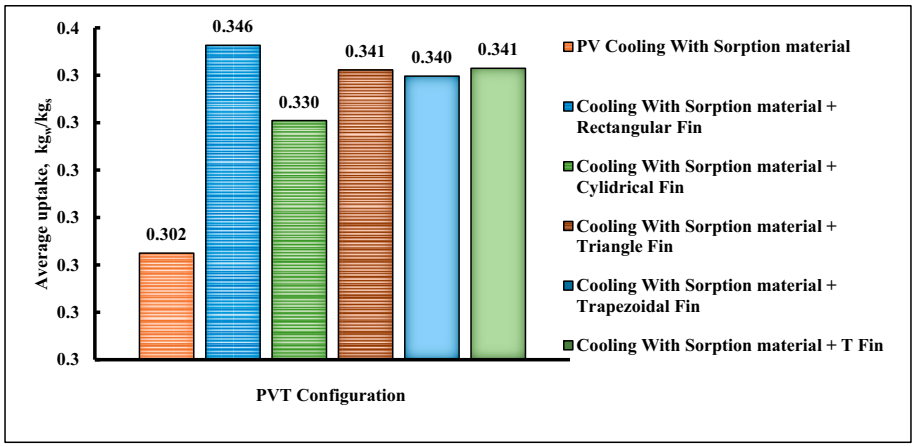


Fig. 13. Average uptake values for each cooling configuration.

may also saturate faster, requiring more frequent regeneration. This trade-off suggests that the optimal design depends on whether peak cooling or longer-duration performance is prioritized.

Overall, the study highlights that while rectangular fins deliver superior thermal performance, triangular or trapezoidal fins may represent better practical compromises in applications requiring compactness, reduced cost, and longer operating cycles. Future work will explore hybrid configurations (e.g., mixed fin geometries or composite materials) to balance performance and feasibility.

7. Conclusions and future recommendations

This study investigated the integration of optimized Silica Gel–$CaCl_2$ sorption material with five fin geometries (rectangular, cylindrical, triangular, trapezoidal, and T-shaped) for passive PV cooling under real climate conditions in Cairo, Egypt. The key findings are summarized as follows:

Thermal performance:

- The uncooled PV module reached a peak of **66.9 °C**, while sorption-only cooling reduced this to **56.0 °C**.
- Hybrid systems achieved even greater reductions, with **rectangular fins lowering peak temperature to 38.6 °C**—a **28.3 °C decrease** compared to the uncooled case.
- Average PV temperatures followed a similar hierarchy: **54.01 °C (uncooled) > 42.89 °C (sorption-only) > 37.75 °C (rectangular fins)**.

Efficiency improvement:

- The uncooled system yielded an average efficiency of **13.03 %**, which improved to **13.78 %** with sorption-only cooling.
- Rectangular fins delivered the highest efficiency (**14.13 %**), corresponding to an **8.4 % relative gain** compared to the baseline.
- Triangular and T-shaped fins performed comparably (14.11 %), while cylindrical fins offered modest improvements (13.99 %).

Moisture uptake performance:

- Sorption-only cooling achieved an average uptake of **0.302**, while fin integration enhanced values to as high as **0.346** (rectangular fins).
- Triangular and T-shaped fins yielded **0.341**, trapezoidal fins **0.340**, and cylindrical fins **0.330**, reinforcing the role of geometry in enhancing desorption-driven cooling.

Unique insights:

- This work establishes, for the first time, a **comparative ranking of fin geometries** in sorption-based PV cooling, showing that **rectangular > triangular ≈ T-shaped > trapezoidal > cylindrical**.
- The results confirm that fins not only enhance conduction pathways but also intensify the sorption–desorption cycle, linking geometry to both thermal and mass transfer performance.
- The validated Multiphysics COMSOL framework offers a transferable tool for designing hybrid cooling systems in hot arid climates.

Future research directions:

- Exploration of **next-generation sorption composites** with higher uptake capacity and improved thermal conductivity.
- Long-term **durability studies** to evaluate cyclic regeneration and retention of cooling capacity.
- **Hybridization strategies** that combine sorption–fin passive cooling with PCMs or liquid-based active methods for enhanced performance.
- Application of the model to **different climatic zones** to generalize the design guidelines.

In summary, the study demonstrates that coupling optimized sorption materials with geometry-tailored fins provides a **quantitatively verified pathway** to reduce PV overheating and improve efficiency in extreme climates. The findings extend beyond incremental gains by offering **actionable design guidance** and a validated computational framework for future PV cooling technologies.

CRediT authorship contribution statement

Moataz M. Abdel-Aziz: Writing – original draft, Visualization, Validation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Asmaa A. ElBahloul:** Writing – review & editing, Supervision, Software, Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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